Year 8

REVISION PACK

Unit 2

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Unit 2 · Solving equations

Q	Working	Answer
one-step and two-step equations		
1.1	**(a)** $q = 13 - 5$ **(b)** $s = 6 \times 4$ **(c)** $15 - 3 = 4y$ $12 = 4y$ $y = 3$ **(d)** $3w = 18 \times 4 = 72$ $w = 72 \div 3 = 24$	**(a)** $q = 8$ **(b)** $s = 24$ **(c)** $y = 3$ **(d)** $w = 24$
1.2	**(a)** $q = 19 - 7$ **(b)** $s = 6 \times 3$ **(c)** $25 - 10 = 5y$ $15 = 5y$ $y = 3$ **(d)** $5w = 15 \times 2 = 30$ $w = 30 \div 5 = 6$	**(a)** $q = 12$ **(b)** $s = 18$ **(c)** $y = 3$ **(d)** $w = 6$
1.3	**(a)** $q = 14 - 9$ **(b)** $s = 2 \times 6$ **(c)** $30 - 6 = 6y$ $24 = 6y$ $y = 4$ **(d)** $3w = 9 \times 5 = 45$ $w = 45 \div 3 = 15$	**(a)** $q = 5$ **(b)** $s = 12$ **(c)** $y = 4$ **(d)** $w = 15$

1.4	**(a)** $q = 30 - 12$ **(b)** $s = 9 \times 2$ **(c)** $17 - 5 = 2y$ $12 = 2y$ $y = 6$ **(d)** $4w = 16 \times 3 = 48$ $w = 48 \div 4 = 12$	**(a)** $q = 18$ **(b)** $s = 18$ **(c)** $y = 6$ **(d)** $w = 12$
1.5	**(a)** $q = 21 - 3$ **(b)** $s = 5 \times 8$ **(c)** $40 - 12 = 7y$ $28 = 7y$ $y = 4$ **(d)** $2w = 6 \times 7 = 42$ $w = 42 \div 2 = 21$	**(a)** $q = 18$ **(b)** $s = 40$ **(c)** $y = 4$ **(d)** $w = 21$
1.6	**(a)** $q = 27 - 15$ **(b)** $s = 4 \times 7$ **(c)** $22 - 7 = 3y$ $15 = 3y$ $y = 5$ **(d)** $7w = 14 \times 2 = 28$ $w = 28 \div 7 = 4$	**(a)** $q = 12$ **(b)** $s = 28$ **(c)** $y = 5$ **(d)** $w = 4$

Unit 2 · Angles and equations

Q	Working	Answer
angles on a straight line		
2.1	$2x + x + 12 = 180$ $3x + 12 = 180$ $3x = 168$ $x = 56$ angle ABD $= 2 \times 56 = 112$	**112°**

2.2	$3x + 20 + 2x + 10 = 180$ $5x + 30 = 180$ $5x = 150$ $x = 30$ $\text{angle DBC} = 2 \times 30 + 10 = 70$	**70°**
2.3	$4x - 5 + x + 35 = 180$ $5x + 30 = 180$ $5x = 150$ $x = 30$ $\text{angle ABD} = 4 \times 30 - 5 = 115$	**115°**
2.4	$5x + 15 + 3x + 5 = 180$ $8x + 20 = 180$ $8x = 160$ $x = 20$ $\text{angle DBC} = 3 \times 20 + 5 = 65$	**65°**
2.5	$2x + 25 + 3x + 5 = 180$ $5x + 30 = 180$ $5x = 150$ $x = 30$ $\text{angle ABD} = 2 \times 30 + 25 = 85$	**85°**
2.6	$6x - 12 + x + 17 = 180$ $7x + 5 = 180$ $7x = 175$ $x = 25$ $\text{angle DBC} = 1 \times 25 + 17 = 42$	**42°**

Unit 2 · Forming equations

Q	Working	Answer
writing and solving an equation from words		
3.1	**(a)** $y + (y + 2) + 2(y + 2) = 62$ **(b)** $4y = 56$ $y = 14$ $Liz = 16,\ James = 2 \times 16 = 32$	**(a)** $4y + 6 = 62$ **(b) 32 years**

3.2	**(a)** $y + (y + 4) + 2(y + 4) = 72$ **(b)** $4y = 60$ $y = 15$ $Noor = 19,\ Adam = 2 \times 19 = 38$	**(a)** $4y + 12 = 72$ **(b) 38 years**
3.3	**(a)** $y + (y - 2) + 3(y - 2) = 42$ **(b)** $5y = 50$ $y = 10$ $Rana = 8,\ Tariq = 3 \times 8 = 24$	**(a)** $5y - 8 = 42$ **(b) 24 years**
3.4	**(a)** $y + (y + 5) + 3(y + 5) = 75$ **(b)** $5y = 55$ $y = 11$ $Sami = 16,\ Yusuf = 3 \times 16 = 48$	**(a)** $5y + 20 = 75$ **(b) 48 years**
3.5	**(a)** $y + (y - 3) + 2(y - 3) = 35$ **(b)** $4y = 44$ $y = 11$ $Zaid = 8,\ Huda = 2 \times 8 = 16$	**(a)** $4y - 9 = 35$ **(b) 16 years**
3.6	**(a)** $y + (y + 6) + 2(y + 6) = 78$ **(b)** $4y = 60$ $y = 15$ $Dina = 21,\ Karim = 2 \times 21 = 42$	**(a)** $4y + 18 = 78$ **(b) 42 years**

Unit 2 · Writing formulae

Q	Working	Answer
writing and using a formula		
4.1	**(a)** **(b)** $974 - 524 = 450$ $450 \div 50 = 9$	**(a)** $T = p + 50n$ **(b) 9 weeks**
4.2	**(a)** **(b)** $455 - 215 = 240$ $240 \div 30 = 8$	**(a)** $T = p + 30n$ **(b) 8 weeks**

4.3	**(a)** **(b)** $640 - 340 = 300$ $300 \div 25 = 12$	**(a)** $T = p + 25n$ **(b) 12 weeks**
4.4	**(a)** **(b)** $700 - 180 = 520$ $520 \div 40 = 13$	**(a)** $T = p + 40n$ **(b) 13 weeks**
4.5	**(a)** **(b)** $257 - 92 = 165$ $165 \div 15 = 11$	**(a)** $T = p + 15n$ **(b) 11 weeks**
4.6	**(a)** **(b)** $1010 - 410 = 600$ $600 \div 60 = 10$	**(a)** $T = p + 60n$ **(b) 10 weeks**

Unit 2 · Solving harder equations

Q	Working	Answer
unknowns on both sides and brackets		
5.1	**(a)** $2x + 9 = 15$ $2x = 6$ $x = 3$ **(b)** $34 - 4y = 5y + 25$ $34 - 25 = 9y$ $9 = 9y$ $y = 1$ **(c)** $7t - 28 = t + 2$ $6t = 30$ $t = 5$	**(a)** $x = 3$ **(b)** $y = 1$ **(c)** $t = 5$

5.2	**(a)** $4x + 1 = 29$ $4x = 28$ $x = 7$ **(b)** $20 - 2y = 3y + 15$ $20 - 15 = 5y$ $5 = 5y$ $y = 1$ **(c)** $5t - 15 = t + 9$ $4t = 24$ $t = 6$	**(a)** $x = 7$ **(b)** $y = 1$ **(c)** $t = 6$
5.3	**(a)** $4x + 4 = 20$ $4x = 16$ $x = 4$ **(b)** $32 - 2y = 4y + 20$ $32 - 20 = 6y$ $12 = 6y$ $y = 2$ **(c)** $5t - 10 = t + 22$ $4t = 32$ $t = 8$	**(a)** $x = 4$ **(b)** $y = 2$ **(c)** $t = 8$
5.4	**(a)** $5x + 9 = 24$ $5x = 15$ $x = 3$ **(b)** $10 - 2y = 4y + 4$ $10 - 4 = 6y$ $6 = 6y$ $y = 1$ **(c)** $3t - 6 = t + 6$ $2t = 12$ $t = 6$	**(a)** $x = 3$ **(b)** $y = 1$ **(c)** $t = 6$

5.5	**(a)** $6x + 8 = 26$ $6x = 18$ $x = 3$ **(b)** $38 - 2y = 4y + 8$ $38 - 8 = 6y$ $30 = 6y$ $y = 5$ **(c)** $6t - 12 = 3t + 15$ $3t = 27$ $t = 9$	**(a)** $x = 3$ **(b)** $y = 5$ **(c)** $t = 9$
5.6	**(a)** $5x + 4 = 14$ $5x = 10$ $x = 2$ **(b)** $10 - 4y = 2y + 4$ $10 - 4 = 6y$ $6 = 6y$ $y = 1$ **(c)** $3t - 12 = t + 2$ $2t = 14$ $t = 7$	**(a)** $x = 2$ **(b)** $y = 1$ **(c)** $t = 7$

Unit 2 · Equations and shapes

Q	Working	Answer
equal sides of a rectangle		
6.1	**(a)** **(b)** $2x = 8$ $x = 4$ $\text{length} = 23,\ \text{width} = 8$ $2 \times 23 + 2 \times 8 = 62$	**(a)** $5x + 3 = 3x + 11$ **(b) 62 cm**

6.2	**(a)** **(b)** $4x = 12$ $x = 3$ $\text{length} = 19,\ \text{width} = 5$ $2 \times 19 + 2 \times 5 = 48$	**(a)** $6x + 1 = 2x + 13$ **(b) 48 cm**
6.3	**(a)** **(b)** $3x = 12$ $x = 4$ $\text{length} = 31,\ \text{width} = 12$ $2 \times 31 + 2 \times 12 = 86$	**(a)** $7x + 3 = 4x + 15$ **(b) 86 cm**
6.4	**(a)** **(b)** $3x = 15$ $x = 5$ $\text{length} = 25,\ \text{width} = 11$ $2 \times 25 + 2 \times 11 = 72$	**(a)** $4x + 5 = x + 20$ **(b) 72 cm**
6.5	**(a)** **(b)** $4x = 16$ $x = 4$ $\text{length} = 38,\ \text{width} = 8$ $2 \times 38 + 2 \times 8 = 92$	**(a)** $9x + 2 = 5x + 18$ **(b) 92 cm**
6.6	**(a)** **(b)** $5x = 25$ $x = 5$ $\text{length} = 44,\ \text{width} = 8$ $2 \times 44 + 2 \times 8 = 104$	**(a)** $8x + 4 = 3x + 29$ **(b) 104 cm**